2019B F=ma Exam: Problem 21

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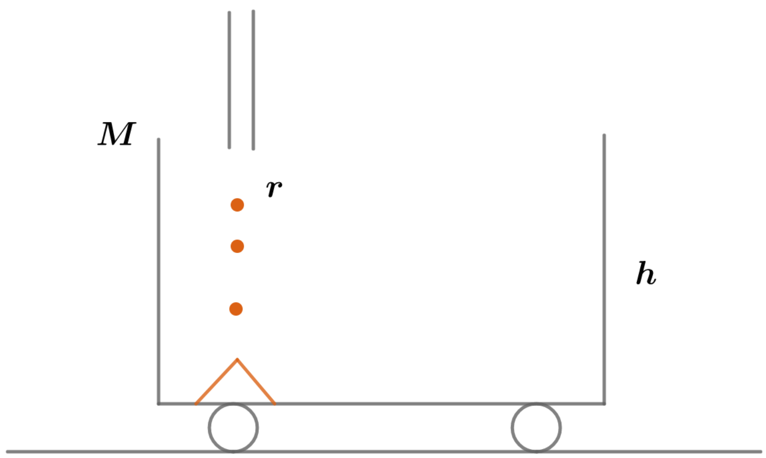

The normal force from the rails goes to support the car's weight, the sand's weight, and to slow down sand landing on the car.

We have

$$F_{car} = Mg$$
$$F_{sand} = rtg$$

By conservation of energy, we have the velocity of the sand landing on the car

$$\frac{1}{2}mv^2 = mgh$$
$$v = \sqrt{2gh}$$

In time Δt, we bring $r\Delta t$ amount of sand to rest. Thus,

$$F_{slow}\Delta t = (r\Delta t)v = r\Delta t\sqrt{2gh}$$
$$F_{slow} = r\sqrt{2gh}$$

Hence,

$$N = F_{car} + F_{sand} + F_{slow} = Mg + rtg + r\sqrt{2gh}$$

so the answer is $\boxed{\text{E}}$.